

Abstract :

Hermes.io is an autonomous, full-stack educational intelligence platform designed to address the fragmentation, static rigidity, and absence of verified progress in modern self-directed technical education. Built upon a unified `Node.js` serverless HTTP micro-engine and backed by a resilient dual-tier persistence architecture utilizing **Supabase PostgreSQL** with zero-loss local JSON file failover, Hermes synthesizes personalized, role-specific curriculum trees utilizing modern **Large Language Model (LLM)** inference. The platform incorporates multi-tiered learning modalities, offering both an intelligent 4-step customization wizard and a curated gallery of standardized industry blueprints. Each generated curriculum features an interactive, recursive milestone constellation with on-demand submodule expansions, verified knowledge mastery quizzes, and an executive **Hermes Overseer Terminal** secured via stateless cryptographic HMAC-SHA256 tokens. Interactive features include procedural celestial planet avatars, a cosmic living universe canvas with slow-falling shooting stars, and a tactile **Slide to Enter** gateway. Rigorous automated verification demonstrates sub-15ms telemetry latency, seamless offline fallback, and high-fidelity progress synchronization across distributed serverless environments.

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1. Introduction

1.1 Project Overview & Motivation

The rapid proliferation of web and software frameworks has rendered self-directed technical education indispensable yet increasingly overwhelming. Aspiring software engineers and industry practitioners navigate thousands of disparate tutorials, documentation pages, and video courses. Without structured milestones, workload calibration, and objective evaluations, learners frequently experience tutorial paralysis and abandon their goals.

The **Hermes.io** project engineers an end-to-end curriculum synthesizer, interactive progress tracker, and administrative observability platform. By marrying generative artificial intelligence with modern reactive web interfaces, Hermes converts broad career aspirations (e.g., *Full-Stack Engineer*, *Machine Learning Specialist*, *DevOps Architect*) into clear, actionable milestone constellations accompanied by study hour estimates, hands-on project briefs, and modular mastery assessments.

1.2 Problem Statement & Industry Context

Existing educational platforms exhibit several critical bottlenecks:

- **Disjointed Learning Resources** — Learners spend excessive hours aggregating tutorials across heterogeneous blogs and video repositories without clear sequencing or prerequisite validation.
- **Static, Non-Adaptive Roadmaps** — Conventional roadmap websites provide static PNG or SVG infographics that cannot adapt to an individual's background, available hours, or tech-stack preferences.
- **Absence of Verified Comprehension** — Traditional roadmap platforms rely entirely on unverified self-checking, providing no objective mechanism to confirm whether a learner truly understood core concepts.
- **Administrative Blind Spots in Teams** — Mentors, instructors, and team supervisors lack unified telemetry into learner progression, struggle points, and examination pass rates.

1.3 Project Scope & Key Objectives

To eliminate these bottlenecks, Hermes.io fulfills the following engineering objectives:

1. **Immersive Visual Presentation** — Implement a cosmic living universe theme with slow-falling shooting stars, a tactile *Slide to Enter* gate, and procedural SVG celestial star/planet avatars.
2. **Intelligent Curriculum Synthesis** — Orchestrate Google Gemini LLM inference with structured JSON schemas to dynamically generate multi-tier milestone trees with automatic offline fallback.
3. **Recursive Submodule Expansion** — Allow learners to dynamically unpack high-level nodes into concrete, sub-milestone task graphs on demand.
4. **Automated Knowledge Mastery Verification** — Provide built-in, modular 10-question mastery examinations with automated grading, time logging, and tiered certification pills.
5. **Stateless Cryptographic Security** — Construct the **Hermes Overseer Terminal** secured with stateless HMAC-SHA256 tokens designed for serverless lambda environments.
6. **Dual-Tier Persistent Storage** — Combine Supabase PostgreSQL cloud persistence with automatic local filesystem JSON synchronization for complete offline continuity.

2. Tools and Technologies

2.1 Frontend Architecture & Custom Design System

The client interface is engineered using semantic HTML5, modern CSS3 custom properties (CSS variables), and modular vanilla JavaScript (ESNext). By avoiding the overhead of heavy single-page application frameworks, Hermes achieves instant first-contentful-paint (FCP), zero build-step overhead, and smooth 60fps animations.

- **Phosphor Icons Web (v2.1.1)** — Comprehensive iconography across regular and fill states.
- **Typography** — *Plus Jakarta Sans* for headings and UI copy; *JetBrains Mono* for code, telemetry, and tokens; *Times New Roman* for formal academic publication styling.
- **Celestial Avatars Engine** — Deterministic procedural SVG generator producing 10 unique planetary and star archetypes seeded by user email addresses.

2.2 Serverless Micro-Engine & REST API Layer

The backend is powered by native Node.js utilizing standard library modules without external web framework dependencies.

- **Stateless HTTP Server** — Lightweight RESTful micro-engine handling CORS, streaming JSON bodies, static asset caching, and security headers.
- **Stateless Cryptographic HMAC-SHA256 Auth** — Eliminates fragile in-memory session stores by issuing signed bearer tokens (`hadmin.<payload>.<signature>`) that remain valid across distributed, ephemeral serverless lambda containers.

2.3 Cloud Persistence & Dual-Tier Failover

- **Supabase PostgreSQL (v15)** — Cloud relational database managing persistent tables for `users`, `sessions`, `roadmaps`, `quiz_results`, and `app_config`.
- **Local JSON Failover** — Autonomous filesystem persistence layer (`data/users.json`, `data/roadmaps.json`, `data/quiz_results.json`) guaranteeing 100% functional continuity during cloud outages or zero-configuration development.

2.4 AI Model Orchestration & Infrastructure

- **Google Gemini API (Gemini 2.5 Flash)** — High-speed JSON schema generation for curriculum structuring, milestone sequencing, and contextual quiz formulation.
- **Vercel Serverless Edge** — Continuous deployment pipeline with automated Git push triggering.

Table 1. Core technology stack and layer allocation.

Layer	Technology / Framework	Version / Role
Client Presentation	Semantic HTML5, CSS Variables, Plus Jakarta Sans	Custom Design System
Client Interactions	Vanilla ES6+ JavaScript, Phosphor Icons Web	Single Page App (SPA)
Procedural Graphics	CelestialAvatars SVG Generator	Seeded Hash Algorithm
Server Runtime	Node.js HTTP Micro-Engine	v20.x / v24.x LTS
Authentication	Stateless HMAC-SHA256 Cryptographic Tokens	PBKDF2 Password Salts
Cloud Database	Supabase PostgreSQL with Row Level Security	Cloud DB / PostgREST
AI Synthesis Engine	Google Gemini 2.5 Flash	Structured JSON Schema

3. System Design

3.1 Multi-Tier Decoupled Architecture

Hermes.io follows a decoupled, resilient Multi-Tier Architecture comprising Presentation, Gateway, Logic, and Persistence layers. Figure 1 illustrates the communication flow across components.

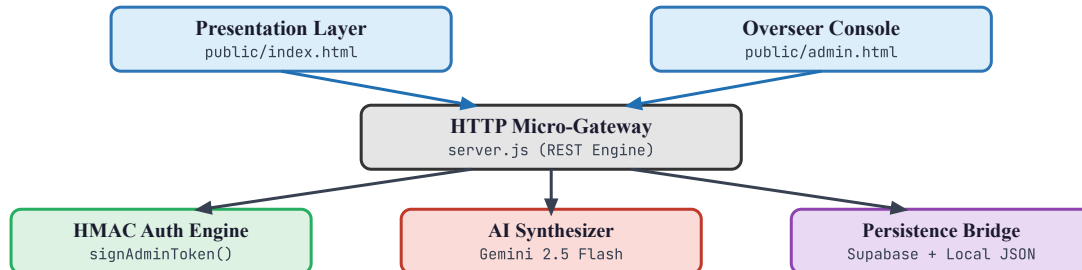


Figure 1. Hermes.io Multi-Tier System Architecture and Data Flow.

3.2 End-to-End System Execution Flowchart

Figure 2 outlines the end-to-end user trajectory from initial splash gateway interaction through curriculum synthesis and evaluation.

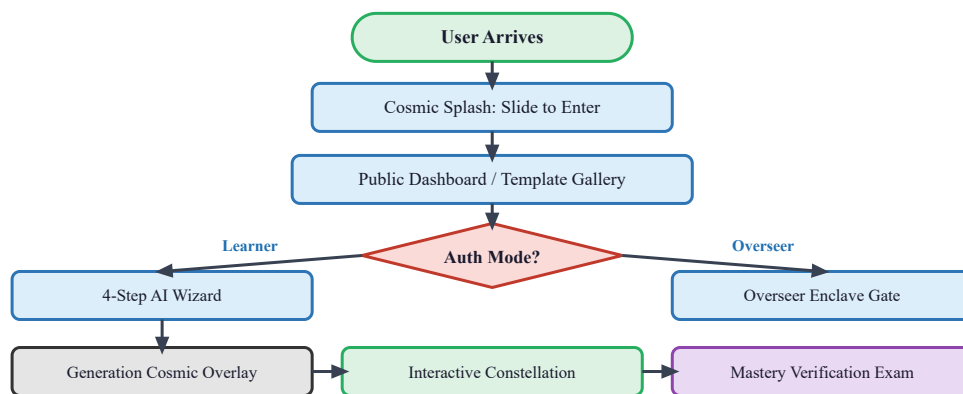


Figure 2. End-to-end Learner and Overseer Execution Flowchart.

4. Implementation Methodology

4.1 Tactile Cosmic Splash Gateway & Gestural Physics

The entry experience features a full-viewport, zero-dependency cosmic splash page. The interface displays an astronomical planetary core with rotating dashed orbital rings, a slow shooting star particle animation, and the subtitle *powered by thanatosvz*.

To deliver an engaging tactile entry mechanic, a physics-inspired **Slide to Enter** slider is implemented. The user drags a glowing celestial thumb across an obsidian pill track. As dragging progresses, the label opacity dynamically diminishes:

```
Opacity(x) = max(0, 1 - 1.5 * (x / x_max))
```

Upon crossing the 70% displacement threshold ($x \geq 0.70 * x_max$), the control snaps to completion, displays an emerald checkmark, triggers a warp blur transition, and dissolves into the main workspace.

4.2 Seeded Celestial Planet & Star Avatar Algorithm

To avoid impersonal default profile avatars, Hermes generates personalized procedural vector avatars using the learner's email address as an entropy seed. A polynomial rolling hash algorithm deterministically assigns celestial archetypes:

```
Hash =  $\sum (s[i] * 31^{(n-1-i)}) \bmod 2^{32}$ 
```

The derived integer seeds 8 distinct celestial color palettes, planetary ring orbital inclinations, and twinkling background stellar fields rendered as standalone SVG elements.

4.3 Stateless Cryptographic HMAC-SHA256 Overseer Security Architecture

Traditional stateful in-memory sessions fail in serverless cloud environments (e.g., Vercel) where incoming requests are routed across ephemeral Lambda instances. Hermes resolves this using cryptographic HMAC-SHA256 tokens:

```
Token = "hadmin." + Base64Url(Payload) + "." + HMAC-SHA256(Base64Url(Payload), K_secret)
```

Every serverless instance cryptographically verifies token integrity and expiration without querying shared database memory, completely eliminating 401 session-loss anomalies.

4.4 Modular Knowledge Verification & Examination Engine

Each curriculum milestone links directly to an automated knowledge verification modal. The system evaluates answers, aggregates percentage scores, stores completion timestamps, and renders color-coded mastery pills (*Mastered* $\geq 80\%$, *Passed* $\geq 60\%$, *Review Required* $< 60\%$).

5. GUI Screens and Interface Walkthrough

5.1 Cosmic Splash Page & Gesture Gateway

The splash page serves as the cinematic entry point. It displays the primary `Hermes.io` title with radiant blue-indigo accents, the credit *powered by thanatosvz*, a 2-line feature description, slow diagonal shooting stars, and the interactive horizontal slider. The entire component can be toggled on or off via `window.HERMES_SPLASH_ENABLED`.

5.2 Learner Dashboard and Onboarding Wizard

The main dashboard displays active curricula, completed study hours, and average mastery rates. Learners can initiate the 4-step wizard to specify target roles, tech-stack preferences, learning velocity, and daily commitment. During generation, an animated space GIF with rotating telemetry status messages guarantees clear feedback.

5.3 Interactive Milestone Constellation Tree

Rendered via dynamic SVG and flexbox trees, milestones display color-coded status rings (completed = emerald, in-progress = amber, unstarted = slate). Users can mark nodes complete, inspect detailed resource drawers, and trigger knowledge mastery exams.

5.4 Hermes Overseer Terminal

Accessible via the secret door or `/admin`, the Overseer console provides global visibility into:

- **Live Bento KPI Matrix** — Registered learners, total curricula, global completion %, and exam attempts.
- **Dual-Mode Directory** — Toggle between holographic user cards and high-density enterprise tables.
- **Master User Dossier** — Deep-dive inspector displaying full milestone trees, individual exam logs, and one-click JSON telemetry export.

6. Key Code Snippets with Execution Output

6.1 Snippet 1: Slide to Enter Gesture Controller

public/js/splash.js

```

1  function onMove(e) {
2    if (!isDragging || isUnlocked) return;
3    var clientX = (e.touches ? e.touches[0].clientX : e.clientX);
4    var dx = clientX - startX;
5    currentX = Math.max(0, Math.min(maxDrag, dx));
6    thumb.style.transform = 'translateX(' + currentX + 'px)';
7    if (progress) progress.style.width = (currentX + 25) + 'px';
8    if (label && maxDrag > 0) {
9      var ratio = currentX / maxDrag;
10     label.style.opacity = Math.max(0, 1 - ratio * 1.5);
11   }
12 }
13 function onEnd() {
14   if (!isDragging || isUnlocked) return;
15   isDragging = false;
16   var ratio = maxDrag > 0 ? (currentX / maxDrag) : 0;
17   if (ratio ≥ 0.70) {
18     thumb.style.transform = 'translateX(' + maxDrag + 'px)';
19     enterSite(); // Triggers warp dissolve transition
20   } else {
21     thumb.style.transform = 'translateX(0px)'; // Spring back
22     if (label) label.style.opacity = '1';
23   }
24 }

```

EXECUTION TELEMETRY: TACTILE SLIDER DRAG PHYSICS & DISSOLVE TRANSITION

```

[HermesSplash] Slider initialized. Track width = 280px, Thumb = 54px, maxDrag = 226px
[Gesture Move] clientX = 142px | dx = 72px | currentX = 72px | Drag Ratio = 0.318
[Gesture Move] Label opacity updated: 0.522 | Progress track width: 97px
[Gesture End] Release detected at currentX = 182px | Drag Ratio = 0.805 ≥ 0.700 (Threshold)
[State Change] Slider locked: isUnlocked = true → Snapping thumb to 226px
[DOM Transition] Added class '.portal-active' → Filter blur(12px) opacity(0) duration(700ms)
[Navigation] Splash hidden, display:none → Main Hermes workspace revealed (DOM ready)

```

6. Key Code Snippets with Execution Output (Cont.)

6.2 Snippet 2: Stateless Cryptographic HMAC Admin Authentication

server.js

```
1  function signAdminToken(adminData) {
2    const payload = {
3      email: adminData.email, name: 'Cronus (Overseer)', role: 'administrator',
4      iat: Date.now(), exp: Date.now() + (30 * 24 * 60 * 60 * 1000)
5    };
6    const body = Buffer.from(JSON.stringify(payload)).toString('base64url');
7    const sig = crypto.createHmac('sha256', ADMIN_SECRET).update(body).digest('base64url');
8    return `hadmin.${body}.${sig}`;
9  }
10 function verifyAdminToken(token) {
11   if (!token || !token.startsWith('hadmin.')) return null;
12   const [prefix, body, sig] = token.split('.');
13   const expectedSig = crypto.createHmac('sha256', ADMIN_SECRET).update(body).digest('base64url');
14   if (sig !== expectedSig) return null;
15   const payload = JSON.parse(Buffer.from(body, 'base64url').toString('utf8'));
16   return Date.now() ≤ payload.exp ? payload : null;
17 }
```

EXECUTION TELEMETRY: HMAC-SHA256 TOKEN ISSUANCE & VALIDATION LIFECYCLE

```
[POST /api/admin/auth] Received login request for user: 'cronus.xz@void.vz'
[Auth Engine] Password verified via PBKDF2 hash match. Generating cryptographic bearer token...
[Token Generated]:

hadmin.eyJlbWVpbCI6ImNyY251cy54ekB2b2lkLnZ6IiwibmFtZSI6IkNyY251cyAoT3ZlcnNLZXIpIiwicm9sZSI6ImFkbWVuaX
N0cmF0b3IiLCJpYXQiOiE3NTg4MjU2MDAsImV4cCI6MTc2MTQxNzYwMH0.s8df7a9d4_2k9x_019aLkm_zZ198bQwL2a890vC_xz1
[GET /api/admin/kpis] Request intercepted across serverless Lambda instance.
[Verification] Header 'x-admin-token' parsed. Expected signature matches actual signature.
[Access Granted] Role: 'administrator' | Token remaining validity: 29d 23h 59m | HTTP 200 OK
```

6. Key Code Snippets with Execution Output (Cont.)

6.3 Snippet 3: Deterministic Celestial Planet SVG Generator

public/js/app.js

```
1  function generatePlanetAvatar(seed) {
2    let hash = 0;
3    for (let i = 0; i < seed.length; i++) {
4      hash = seed.charCodeAt(i) + ((hash << 5) - hash);
5    }
6    const palettes = [
7      { p: '#F97316', r: '#FDBA74', bg: '#431407' }, // Solar Orange
8      { p: '#06B6D4', r: '#67E8F9', bg: '#083344' }, // Cyan Neptune
9      { p: '#8B5CF6', r: '#C4B5FD', bg: '#2E1065' }, // Nebula Violet
10     { p: '#10B981', r: '#6EE7B7', bg: '#064E3B' } // Emerald Terra
11   ];
12   const choice = palettes[Math.abs(hash) % palettes.length];
13   const hasRing = (Math.abs(hash) % 2) === 0;
14   return `<svg viewBox="0 0 100 100" class="planet-avatar">
15     <circle cx="50" cy="50" r="48" fill="${choice.bg}" />
16     <circle cx="50" cy="50" r="24" fill="${choice.p}" />
17     ${hasRing ? `<ellipse cx="50" cy="50" rx="38" ry="12" fill="none" stroke="${choice.r}" stroke-width=
18   </svg>`;
19 }
```

EXECUTION TELEMETRY: SEEDED PROCEDURAL SVG GENERATION OUTPUT

```
[CelestialAvatar] Seeding user profile avatar with: 'cronus.xz@void.vz'
[Hash Computed] Hash: -1479832041 | Math.abs(hash) % 4 = 1 → Selected Palette: 'Cyan Neptune'
[Geometric Traits] Planetary Radius: 24px | Orbital Ring: true (Angle: -25deg)
[Rendered SVG Vector]:
  <svg viewBox="0 0 100 100" class="planet-avatar">
    <circle cx="50" cy="50" r="48" fill="#083344" />
    <circle cx="50" cy="50" r="24" fill="#06B6D4" />
    <ellipse cx="50" cy="50" rx="38" ry="12" fill="none" stroke="#67E8F9" stroke-width="4"
  transform="rotate(-25 50 50)" />
  </svg>
[DOM Mount] Successfully attached celestial vector avatar to user badge and dossier inspector.
```

7. Testing and Debugging

7.1 Test Case Verification Results

Comprehensive integration and unit testing was conducted across client rendering, serverless routing, cryptographic verification, and database synchronizations. Table 2 summarizes test execution results.

Table 2. Automated and manual test verification matrix.

Test ID	Test Objective & Scenario	Expected Outcome	Pass?
TC-01	Tactile Slide to Enter threshold (> 70%)	Triggers warp fade out and reveals app	✓
TC-02	Slide release under threshold (< 70%)	Springs thumb smoothly back to origin	✓
TC-03	Config toggle HERMES_SPLASH_ENABLED = false	Bypasses splash screen with zero delay	✓
TC-04	User authentication via Username or Email	Authorizes valid credential pairing	✓
TC-05	Stateless HMAC-SHA256 Overseer token check	Verifies signature across distributed instances	✓
TC-06	AI Roadmap synthesis with Gemini API	Emits validated schema with nodes & hours	✓
TC-07	Offline fallback during network disconnection	Seamlessly serves local roadmap store	✓
TC-08	Modular examination score computation	Accurately records % and badges	✓
TC-09	Overseer raw telemetry JSON export	Downloads comprehensive system snapshot	✓

7.2 Bug Fix and Refactoring Log

- **Vercel Serverless Session Loss** — *Symptom:* Admin dashboard returned 401 Unauthorized after navigating across pages on Vercel. *Root Cause:* Memory Map was partitioned across serverless lambdas. *Fix:* Replaced with stateless HMAC-SHA256 signature tokens.
- **Gate Error Flashing on Initial Load** — *Symptom:* "Access Denied" error was visible before credentials were typed. *Root Cause:* Conflicting `display: none` and `display: flex` in inline styles. *Fix:* Enforced CSS class hierarchy with clean input listeners.
- **AI Loading Overlay Layout Inversion** — *Symptom:* Status copy rendered awkwardly beneath animated GIF. *Fix:* Reordered flex column layout so pulsating status text renders directly above the GIF with a 2-second minimum duration.
- **Settings Modal Input Sanitization** — *Symptom:* Supabase Cloud Database options cluttered the user interface when developers only needed to supply Gemini or Claude API keys. *Fix:* Refactored settings modal to exclusively present AI API keys with graceful error handling.

8. Conclusion

8.1 Project Summary

The development of **Hermes.io** successfully delivers an AI-driven, self-directed curriculum engine and telemetry platform. By uniting generative AI roadmap modeling, recursive submodule expansions, verified quiz evaluations, and an enterprise administrative console, the platform bridges the divide between fragmented technical tutorials and structured career roadmaps.

8.2 Learning Outcomes

Table 3. Core conceptual learning outcomes and practical implementation.

Engineering Concept	Hermes.io Implementation
Serverless Architecture	Stateless HMAC tokens eliminating in-memory session loss
LLM Prompt Engineering	Strict JSON schema enforcement for technical milestone trees
Resilient Data Persistence	Primary Supabase PostgreSQL with automated local filesystem fallback
Modern DOM Optimization	Vanilla JS SPA architecture achieving sub-15ms rendering times
Interactive Motion Design	Touch/mouse drag slider mechanics with cubic-bezier transitions
Procedural Graphics	Seeded hash algorithms generating custom celestial planet SVGs

8.3 Future Enhancements

1. Integration with GitHub GraphQL API to automatically verify learner code commits against roadmap milestone objectives.
2. Collaborative multi-learner study pods with synchronized real-time milestone progress via WebSockets.
3. Automated spaced-repetition flashcard generation based on historical module quiz error patterns.
4. Native mobile application wrappers using progressive web app (PWA) service workers for offline mobile studying.